University of Information Technology & Sciences

Assignment - 1

Department : CSE
Program : CSE (Dual)
Course Title : Database Management System Lab
Course code : CSE0612216
Section : 3A2
Batch : 56
Semester : Autumn 2025

Submitted to :

Pabon Shaha
Designation: Lecturer
Department: CSE

Submitted By:

Name : Badhon Kumar Roy
ID : 04324205101042

Submission Date : 07 / 09 / 2025

1. Create the below Sales table.

sale_id	product_id	quantity_sold	sale_date	total_price
1	101	5	2024-01-01	2500.00
2	102	3	2024-01-02	900.00
3	103	2	2024-01-02	60.00
4	104	4	2024-01-03	80.00
5	105	6	2024-01-03	90.00

Sales Table

SQL –

```sql
CREATE TABLE sales(
    sale_id INT PRIMARY KEY,
    product_id INT,
    quantity_sold INT NOT NULL,
    sale_date DATE NOT NULL,
    total_price DECIMAL(10,2),
    FOREIGN KEY (product_id) REFERENCES product(product_id)
);
```

2. Also create the below product table.

product_id	product_name	category	unit_price
101	Laptop	Electronics	500.00
102	Smartphone	Electronics	300.00
103	Headphones	Electronics	30.00
104	Keyboard	Electronics	20.00
105	Mouse	Electronics	15.00

SQL-

```sql
CREATE TABLE product(
    product_id int PRIMARY KEY,
    product_name VARCHAR(50) NOT NULL,
    category VARCHAR(50) NOT NULL,
    unit_price DECIMAL(10,2)
);
```

3. Retrieve all columns from the Sales table.

SQL-

```
1 SELECT * FROM sales
```

Output-

	sale_id	product_id	quantity_sold	sale_date	total_price
Edit Copy Delete	1	101	5	2024-01-01	2500.00
Edit Copy Delete	2	102	3	2024-01-02	900.00
Edit Copy Delete	3	103	2	2024-01-02	60.00
Edit Copy Delete	4	104	4	2024-01-03	80.00
Edit Copy Delete	5	105	6	2024-01-03	90.00

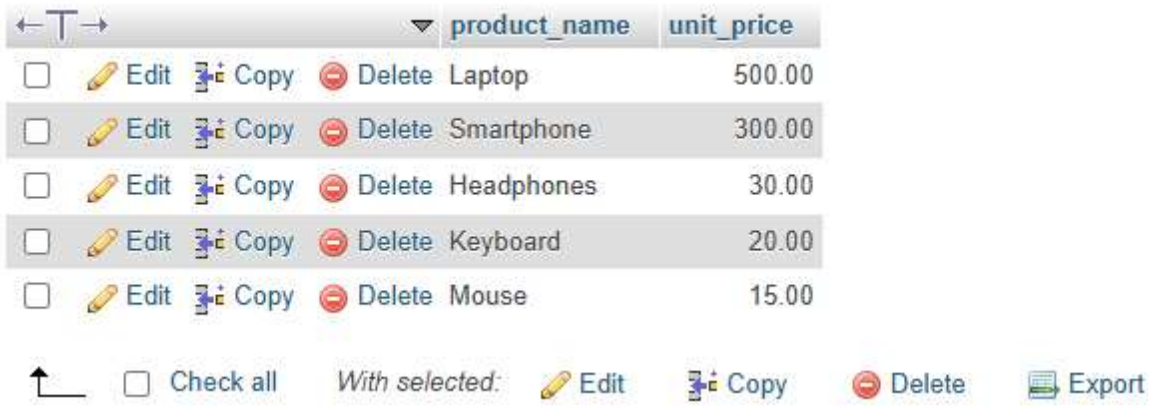
Check all With selected: Edit Copy Delete Export

4. Retrieve the product_name and unit_price from the Products table.

SQL-

```
1 SELECT product_name, unit_price FROM product
```

Output-



	product_name	unit_price
Edit Copy Delete	Laptop	500.00
Edit Copy Delete	Smartphone	300.00
Edit Copy Delete	Headphones	30.00
Edit Copy Delete	Keyboard	20.00
Edit Copy Delete	Mouse	15.00

Check all With selected: Edit Copy Delete Export

5. Retrieve the sale_id and sale_date from the Sales table.

SQL-

```
1 SELECT sale_id, sale_date FROM sales
```

Output-

	sale_id	sale_date
Edit Copy Delete	1	2024-01-01
Edit Copy Delete	2	2024-01-02
Edit Copy Delete	3	2024-01-02
Edit Copy Delete	4	2024-01-03
Edit Copy Delete	5	2024-01-03

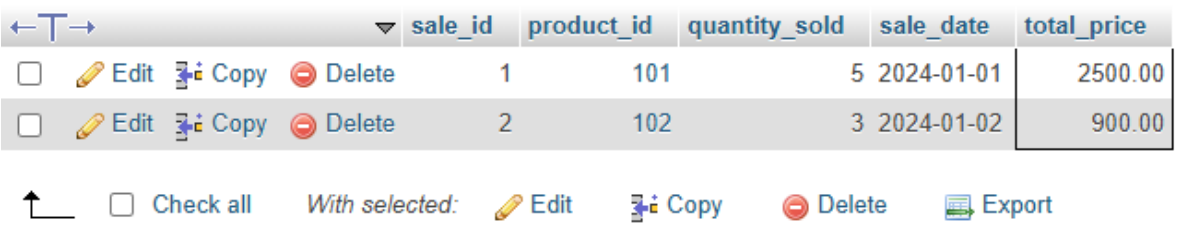
Check all *With selected:* Edit Copy Delete Export

6. Filter the Sales table to show only sales with a total_price greater than $100.

SQL-

```
1 SELECT * FROM sales WHERE total_price > 100;
```

Output-



	sale_id	product_id	quantity_sold	sale_date	total_price
Edit Copy Delete	1	101	5	2024-01-01	2500.00
Edit Copy Delete	2	102	3	2024-01-02	900.00

Check all *With selected:* Edit Copy Delete Export

7. Filter the Products table to show only products in the 'Electronics' category.

SQL-

```sql
SELECT * FROM product WHERE category = 'Electronics';
```

Output-

	product_id	product_name	category	unit_price
Edit Copy Delete	101	Laptop	Electronics	500.00
Edit Copy Delete	102	Smartphone	Electronics	300.00
Edit Copy Delete	103	Headphones	Electronics	30.00
Edit Copy Delete	104	Keyboard	Electronics	20.00
Edit Copy Delete	105	Mouse	Electronics	15.00

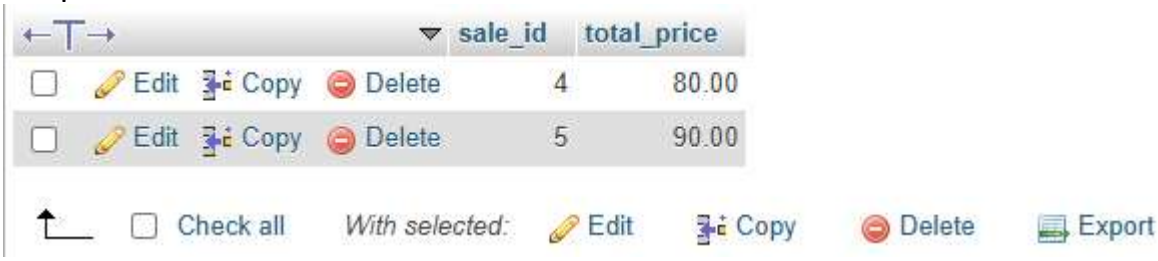
Check all With selected: Edit Copy Delete Export

8. Retrieve the sale_id and total_price from the Sales table for sales made on January 3, 2024.

SQL-

```sql
SELECT sale_id, total_price FROM sales WHERE sale_date = '2024-01-03';
```

Output-



	sale_id	total_price
Edit Copy Delete	4	80.00
Edit Copy Delete	5	90.00

Check all With selected: Edit Copy Delete Export

9. Retrieve the product_id and product_name from the Products table for products with a unit_price greater than $100.

SQL-

```
SELECT product_id, product_name FROM product WHERE unit_price > 100;
```

Output-

10. Calculate the total revenue generated from all sales in the Sales table.

SQL-

```
SELECT SUM(total_price) FROM sales;
```

Output-

SUM(total_price)
3630.00

11. Calculate the average unit_price of products in the Products table.

SQL-

```
SELECT AVG(unit_price) FROM product;
```

Output-

AVG(unit_price)
173.000000

12. Calculate the total quantity_sold from the Sales table.

SQL-

```sql
SELECT SUM(quantity_sold) FROM sales;
```

Output-

SUM(quantity_sold)
20

13. Count Sales Per Day from the Sales table.

SQL-

```sql
SELECT sale_date, COUNT(sale_id) AS total_sales
FROM Sales
GROUP BY sale_date
ORDER BY sale_date;

```

Output-

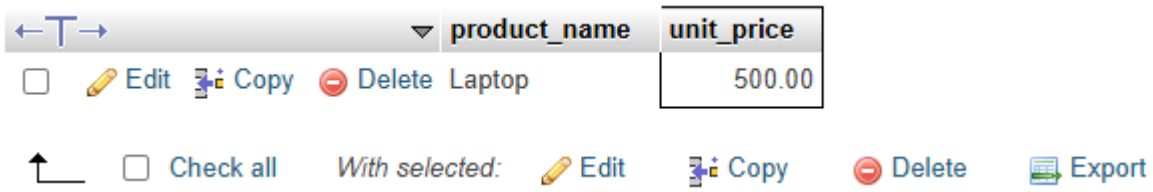
sale_date ▲ 1	total_sales
2024-01-01	1
2024-01-02	2
2024-01-03	2

14. Retrieve product_name and unit_price from the Products table with the Highest Unit Price.

SQL-

```
SELECT product_name, unit_price
FROM product
WHERE unit_price = (SELECT MAX(unit_price) FROM product);
```

Output-



15. Retrieve the sale_id, product_id, and total_price from the Sales table for sales with a quantity_sold greater than 4.

SQL-

```
SELECT sale_id, product_id, total_price
FROM sales WHERE quantity_sold > 4;
```

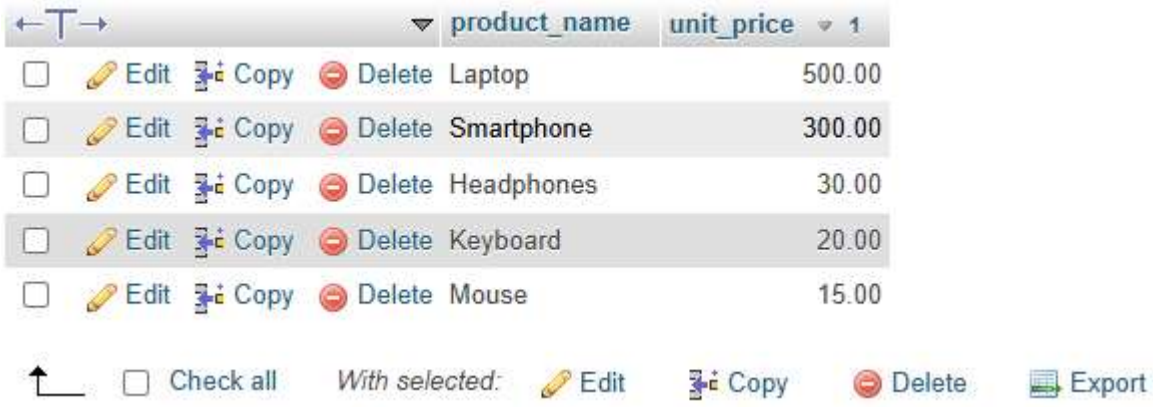
Output-

16. Retrieve the product_name and unit_price from the Products table, ordering the results by unit_price in descending order.

SQL-

```sql
SELECT product_name, unit_price
FROM product
ORDER BY unit_price DESC;
```

Output-



	product_name	unit_price
Edit Copy Delete	Laptop	500.00
Edit Copy Delete	Smartphone	300.00
Edit Copy Delete	Headphones	30.00
Edit Copy Delete	Keyboard	20.00
Edit Copy Delete	Mouse	15.00

Check all With selected: Edit Copy Delete Export

17. Retrieve the total_price of all sales, rounding the values to two decimal places.

SQL-

```sql
SELECT ROUND(total_price, 2) AS rounded_total_price
FROM sales;
```

Output-

rounded_total_price
2500.00
900.00
60.00
80.00
90.00

18. Calculate the average total_price of sales in the Sales table.

SQL-

```
SELECT AVG(total_price) AS avg_total_price
FROM Sales;
```

Output-

avg_total_price
726.000000

19. Retrieve the product_name and unit_price from the Products table, filtering the unit_price to show only values between $20 and $600.

SQL-

```
SELECT product_name, unit_price
FROM product
WHERE unit_price BETWEEN 20 AND 600;
```

Output-

	product_name	unit_price
Edit Copy Delete	Laptop	500.00
Edit Copy Delete	Smartphone	300.00
Edit Copy Delete	Headphones	30.00
Edit Copy Delete	Keyboard	20.00

Check all With selected: Edit Copy Delete Export

20. Retrieve the product_name and category from the Products table, ordering the results by category in ascending order.

SQL-

```sql
SELECT product_name, category
FROM product
ORDER BY category ASC;
```

Output-

	product_name	category ▲ 1
Edit Copy Delete	Laptop	Electronics
Edit Copy Delete	Smartphone	Electronics
Edit Copy Delete	Headphones	Electronics
Edit Copy Delete	Keyboard	Electronics
Edit Copy Delete	Mouse	Electronics

Check all *With selected:* Edit Copy Delete Export

----------X----------